

Introduction to Interactive Machine Learning (IML)

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ML & Human(s)

ML & Human(s)

Who is helping Who?

- **Machine helping the human**

Amplifying human abilities

- **Humans helping the machine**

Providing labels, preferences and guidance

- **Towards a collaboration / cooperation / partnership?**

Crandall et al. (2018). Cooperating with machines. Nature Communications, 9(1), 1-12

Tschandl et al. (2020). Human-computer collaboration for skin cancer recognition.

Nature Medicine, 26(8), 1229-1234



A.I. Is Learning From Humans. Many Humans.

Artificial intelligence is being taught by thousands of office workers around the world. It is not exactly futuristic work.

Crowd for Supervised Learning

Human Computation: On-demand and scalable access to human intelligence for tasks that computers cannot yet do alone.

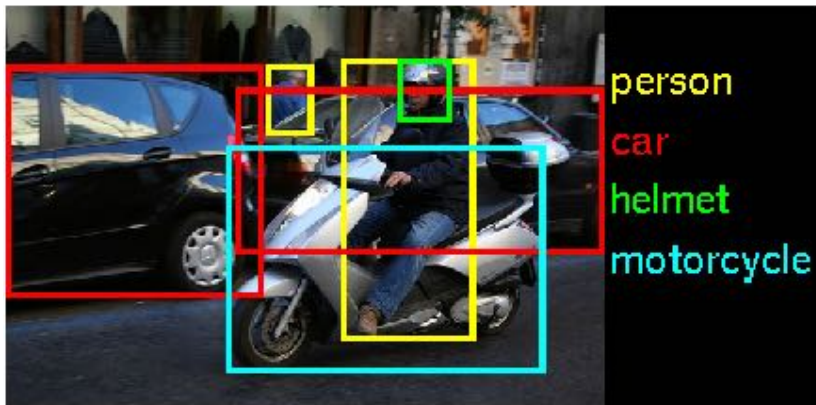
=> *user crowds to annotate (label) massive dataset*

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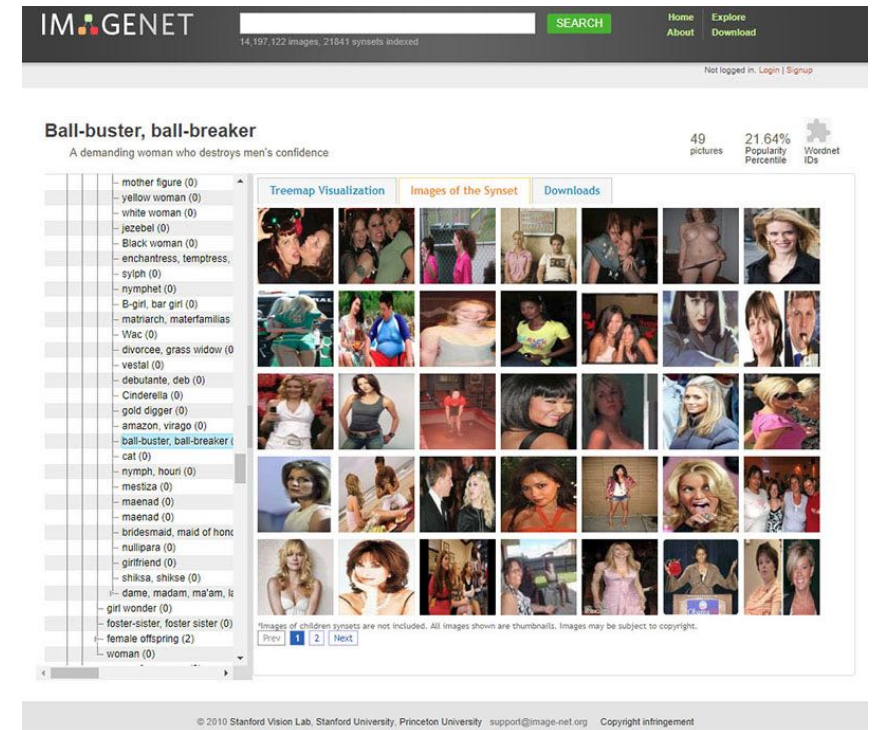
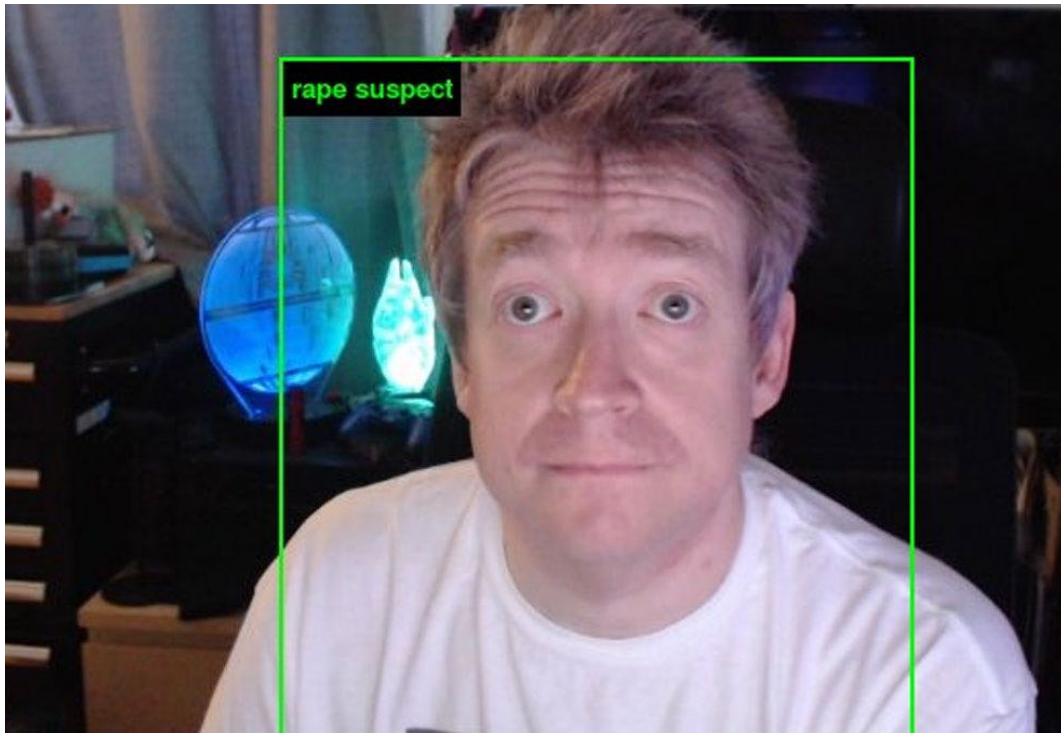
Example: **ImageNet**



- Large visual database for vision research
- Constructed with crowdsourcing
- Total number of images: 14,197,122
- Number of images with bounding box annotations: 1,034,908

Deng, J., Dong, W., Socher, R., Li, L. J., Li, K., & Fei-Fei, L. (2009, June). Imagenet: A large-scale hierarchical image database. In 2009 IEEE conference on computer vision and pattern recognition (pp. 248-255). IEEE.

Human biases => machine biases

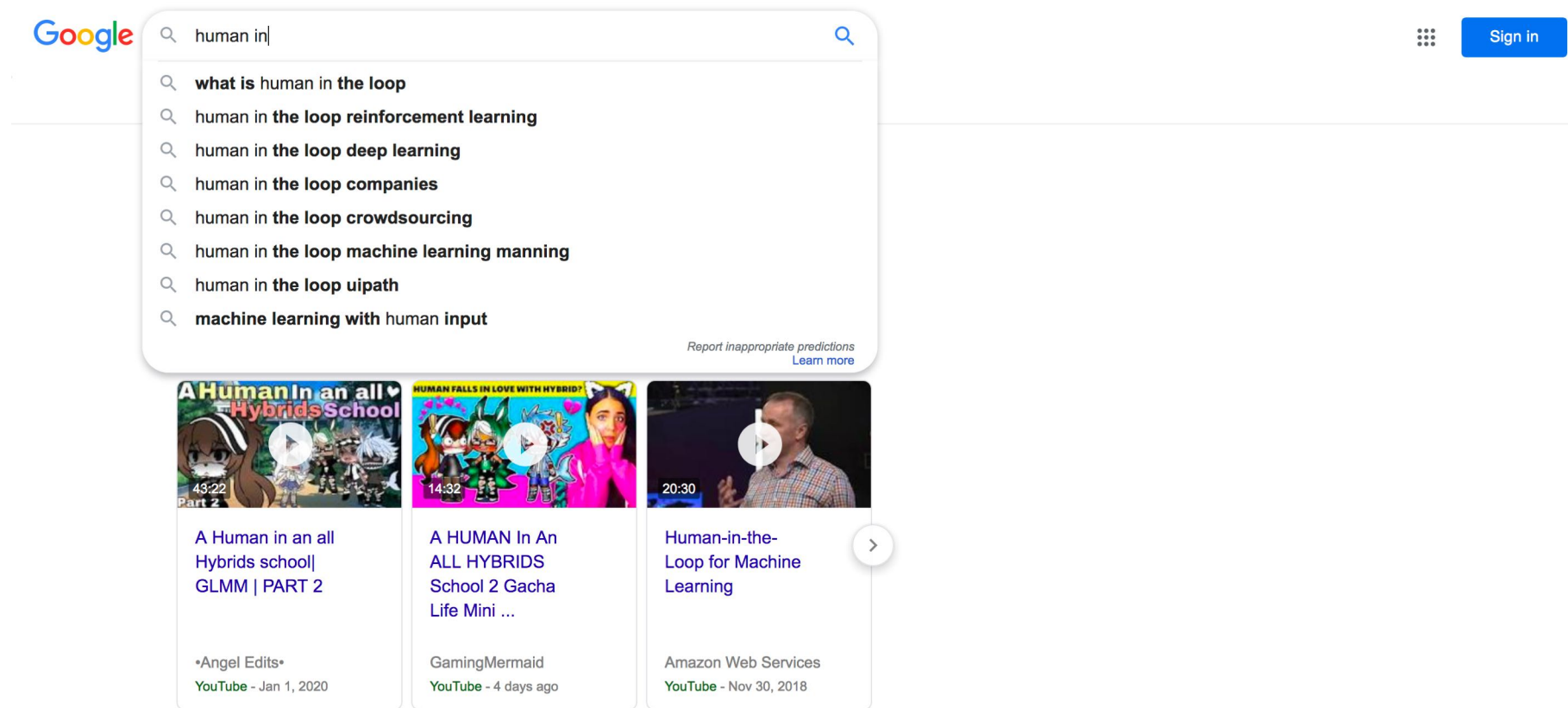


Deng, J., Dong, W., Socher, R., Li, L. J., Li, K., & Fei-Fei, L. (2009, June). Imagenet: A large-scale hierarchical image database. In 2009 IEEE conference on computer vision and pattern recognition (pp. 248-255). IEEE.

Human-in-the-loop ML

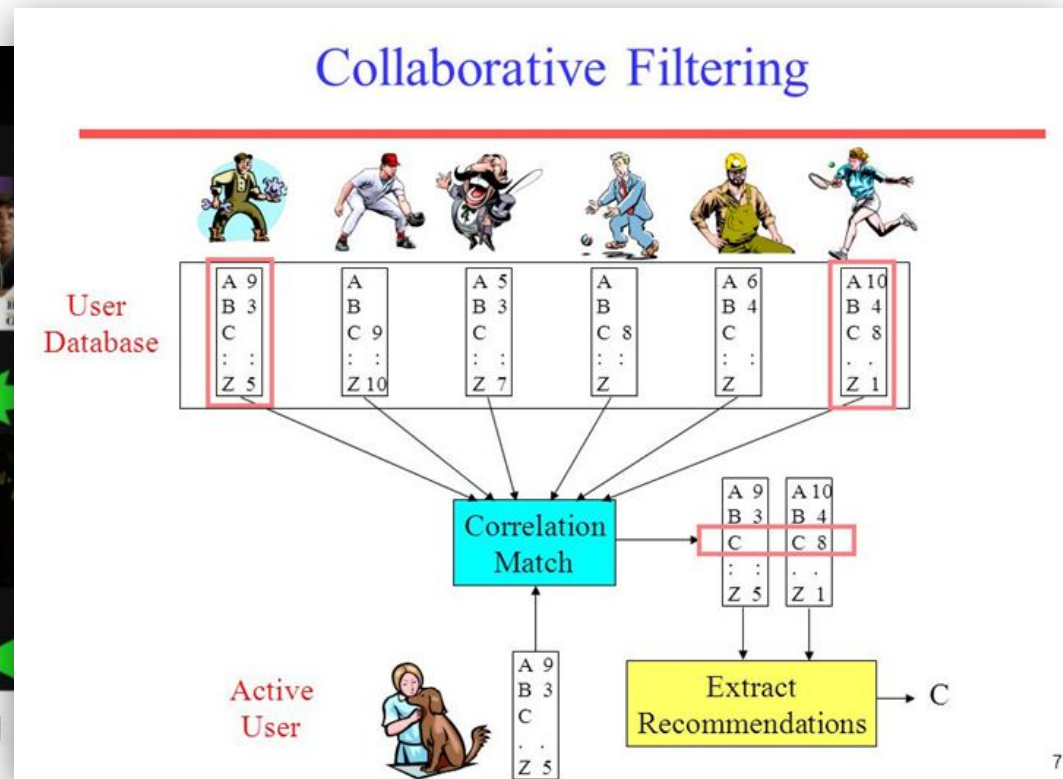
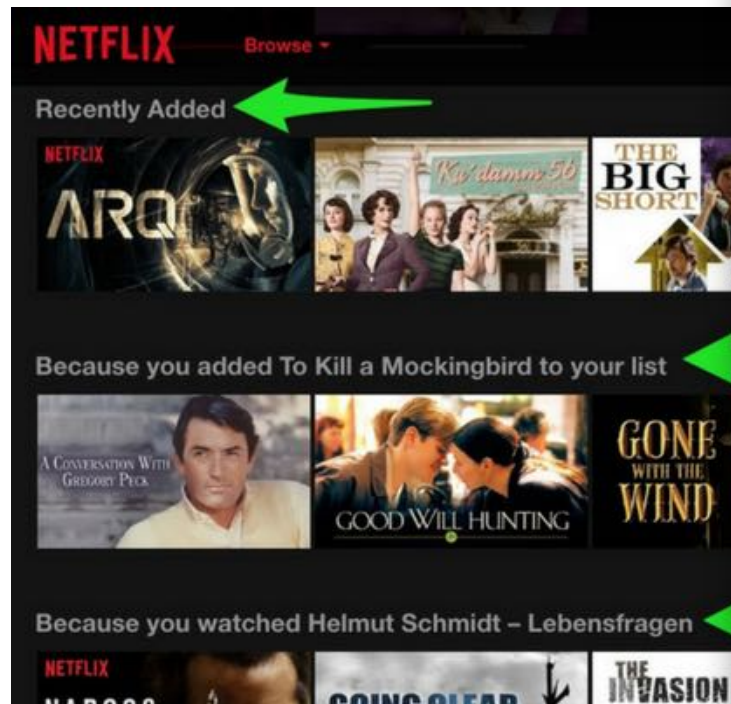
Human-in-the-loop ML

Use human input to **improve the algorithm**



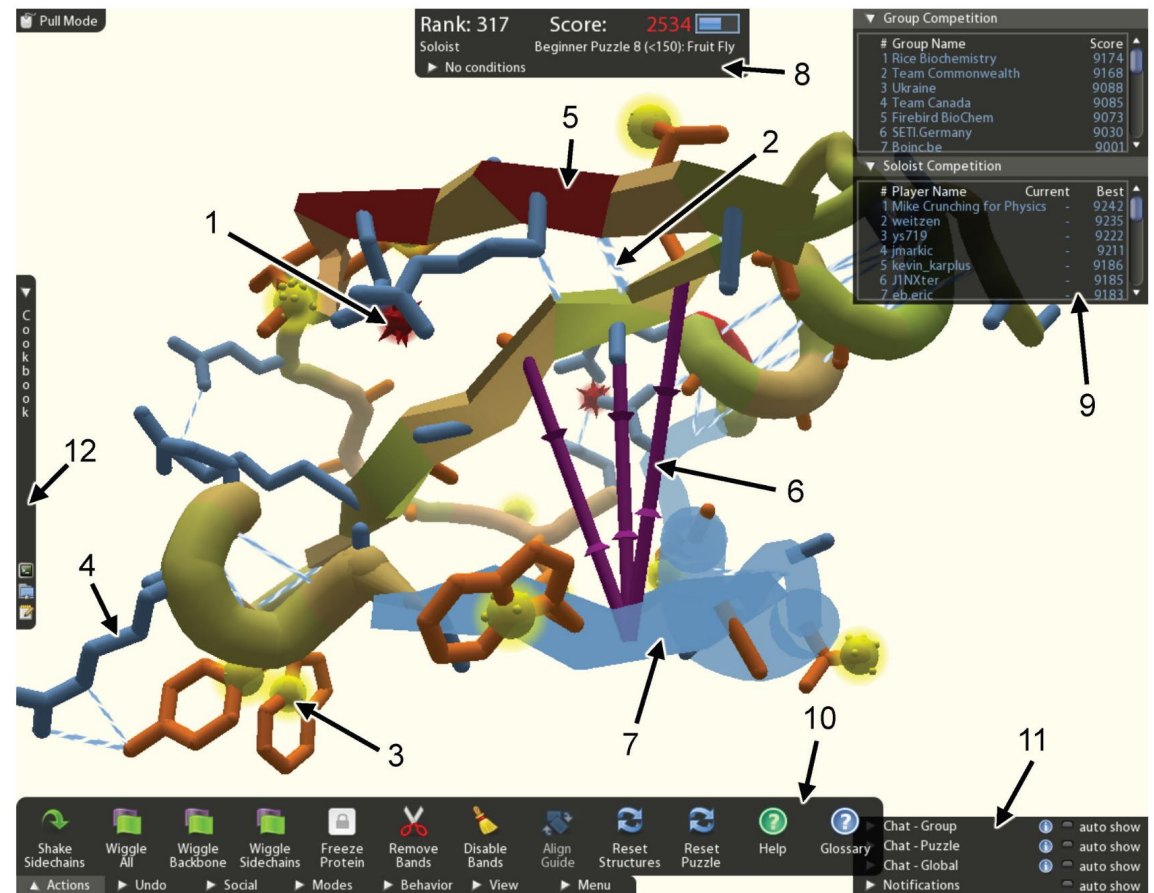
Example: Recommender Systems

Combine human input to **personalise** predictions



Example: Expert Domain

Use crowd intelligence to help
protein structure discovery:
Foldit



Cooper et al. (2010) Predicting protein structures with a multiplayer online game. *Nature* 466(7307):756–760
<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC2956414/pdf/nihms218516.pdf>

Interactive Machine Learning

Interactive Machine Learning

End-user Interaction with the Machine Learning process

« Interactive Machine Learning is an interaction paradigm in which a user or user group iteratively builds and refines a mathematical model to describe a concept through iterative cycles of input and review »

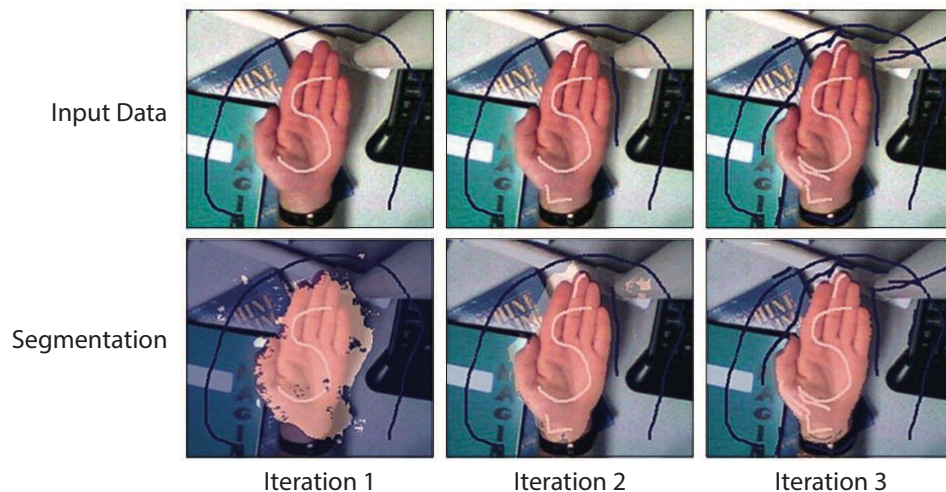
Dudley, J. J., & Kristensson, P. O. (2018). A review of user interface design for interactive machine learning. *ACM Transactions on Interactive Intelligent Systems (TiiS)*, 8(2), 8.

Motivations

- The complexity of machine learning has restricted its use to experts and skilled developers
- Real-world problems can benefit from end-user interaction
- Better understand models
- Better assess a model's performance
- ...

Crayons

- **Task:** image segmentation
- Goal: identify foreground/background in images
- Users can draw to mark if pixels are foreground or background



=> Users can easily identify classification errors and provide more annotations to improve the classifier

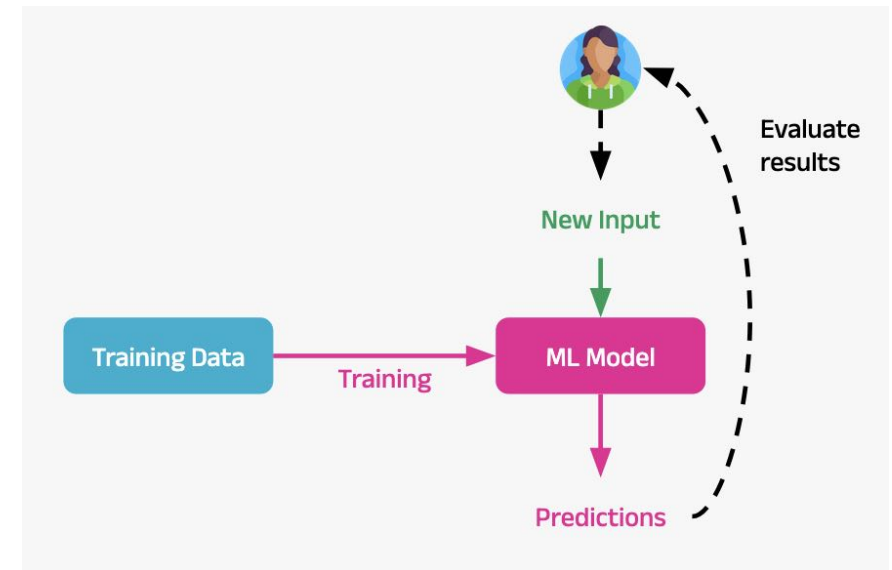
What IML is not

- Considering that **people are only useful** for ML
 - Using people to label large datasets
 - Exploiting user's choices/preferences to improve an algorithm (blindly)
- Using an interactive system **integrating a ML algorithm**
 - Using a computer vision algorithm to track a person's skeleton
 - Integrating speech recognition

Traditional ML workflow

Development (by ML experts):

1. Collect data
2. Select features to represent the data
3. Preprocess and transform the data
4. Choose a representation and learning algorithm to construct the model
5. Tune parameters of the algorithm
6. Assess the quality of the resulting model



Usage (by end-users): limited to providing data/annotations, answering domain-related questions, or giving feedback about the learned model.

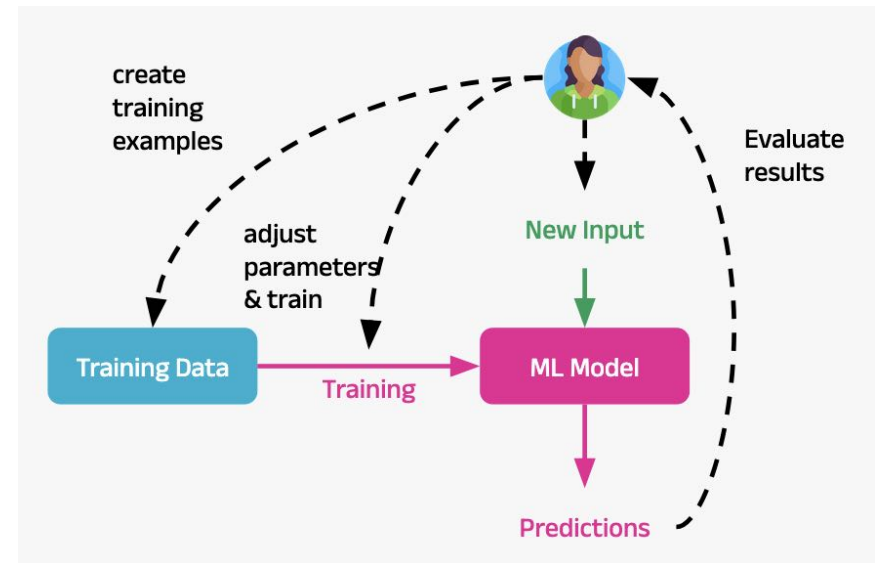
Interactive ML workflow

Users (potentially) involved at all steps of the process

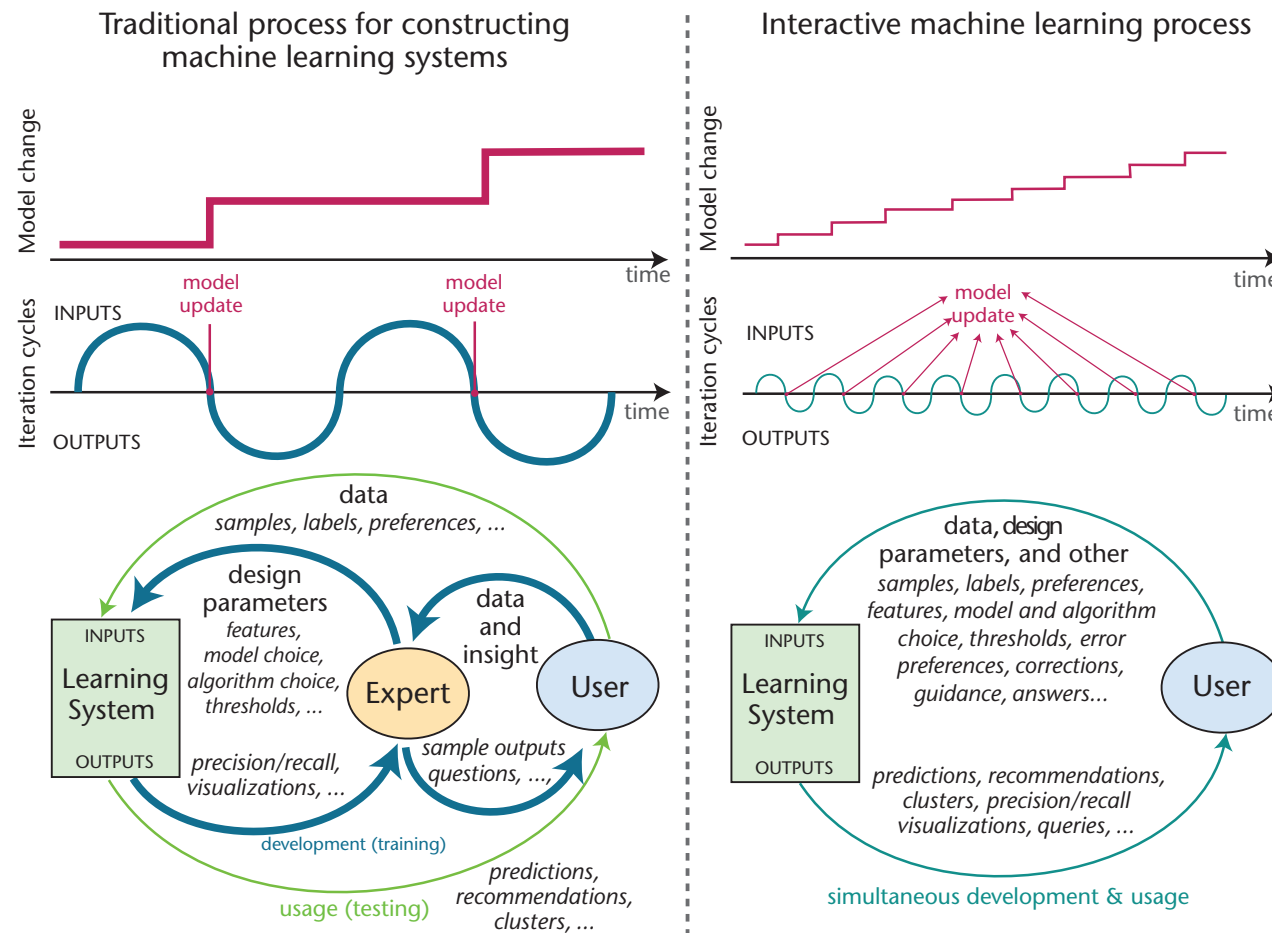
Iterative model refinement

- Driven by user input
- Through various modalities: samples, labels, features, preferences, model choices, ...
- With continuous feedback from the model
- Towards a certain design objective

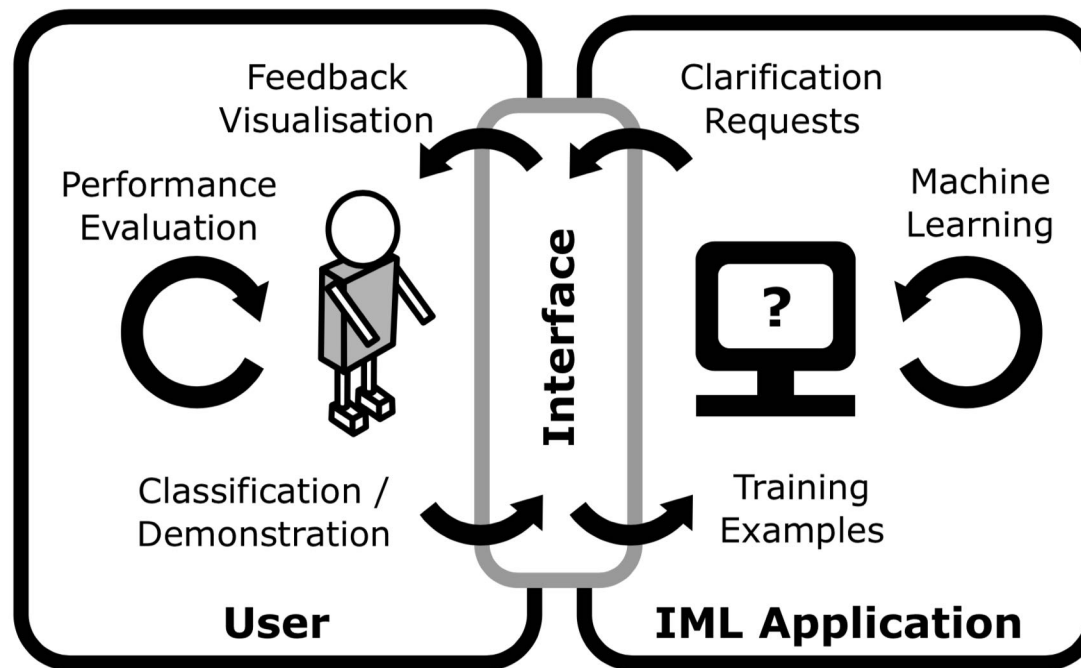
Simultaneous development and usage



Traditional v. IML workflows

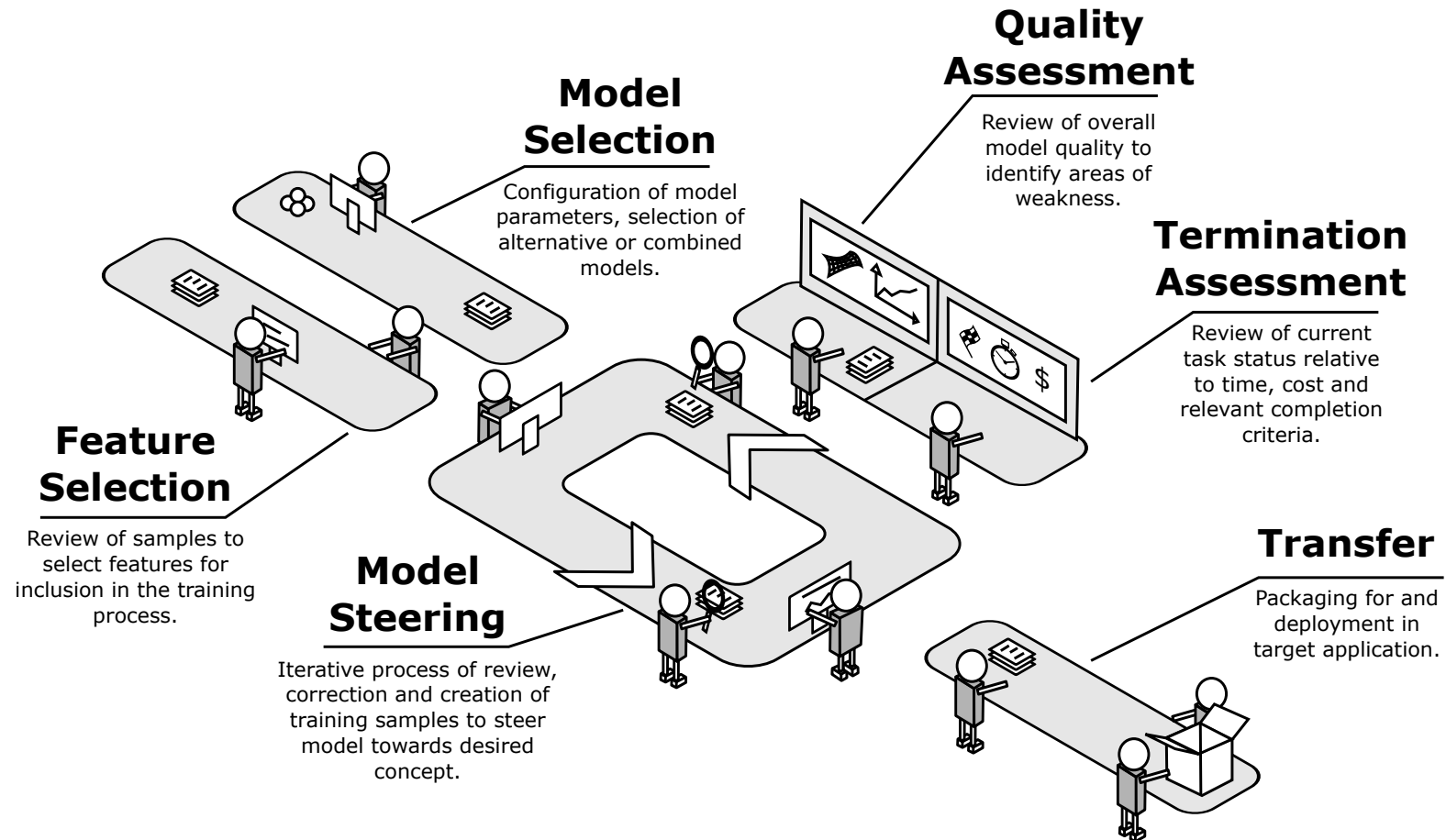


Overview of an IML System



Dudley, J. J., & Kristensson, P. O. (2018). A review of user interface design for interactive machine learning. *ACM Transactions on Interactive Intelligent Systems (TiiS)*, 8(2), 8.

IML Workflow



Dudley, J. J., & Kristensson, P. O. (2018). A review of user interface design for interactive machine learning. *ACM Transactions on Interactive Intelligent Systems (TiiS)*, 8(2), 8.

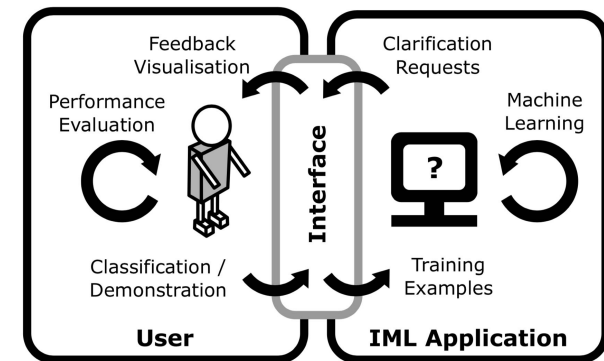
Overview of an IML System

User:

- Main driver of the interaction
- Usually, not a ML expert
- Potentially unreliable (concept drift, variability, ...)

Model:

- The component that makes predictions from input
- Can be parametrised and retrained
- Need to be appropriate for IML (fast training, few examples, ...)



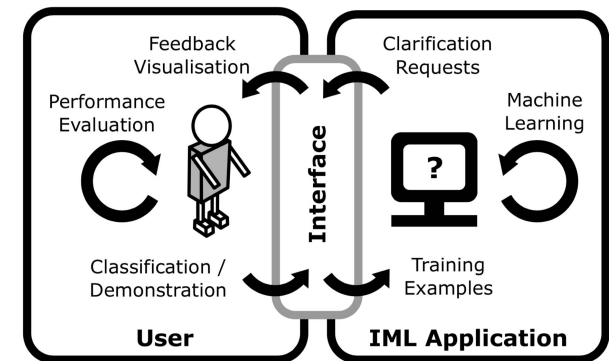
Overview of an IML System

Data:

- Provided and labeled by the user
- Should describe the system's desired behavior
- Variable across applications

Interface:

- bidirectional feedback between the user and the model/data
- Support input+output
- Critical for user's understanding and expressivity



Key design principles

1. Make task goals and constraints explicit
2. Support user understanding of model uncertainty and confidence
3. Capture intent rather than input
4. Provide effective data representations
5. Exploit interactivity and promote rich interactions
6. Engage the user

ACTIVITY

Activity 1

Go to:

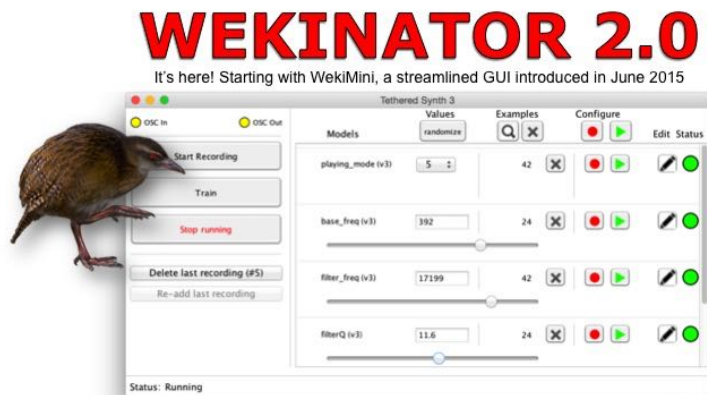
<https://demos.marcelle.dev/sketch-detailed/>

1. Draw and record drawings with labels
2. Train the model
3. Inspect results and make edits
4. Group discussion

IML Examples

Wekinator

- **Task:** Music performance
- Goal: create novel gesture-based instruments
- Regression to learn movement-sound mapping



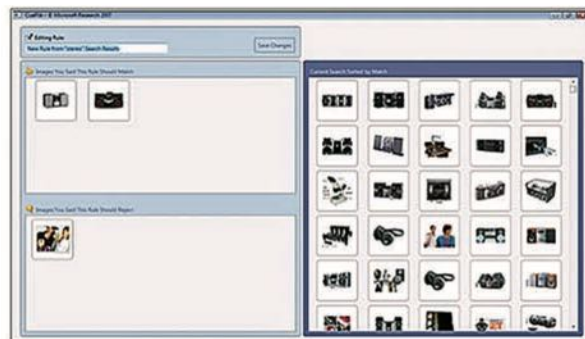
<http://www.wekinator.org/example-projects/>

=> Users learn how to give examples
and update their expectations along
the process

Fiebrink et al. (2011). Human model evaluation in interactive supervised learning. In Proceedings of CHI (pp. 147-156). ACM. [PDF](#)

CueFlik

- **Task:** image classification
- Goal: improve image-based web search
- Allow end-users to build rules by demonstration of pos/neg examples



Standard presentation using
a ranked list of examples



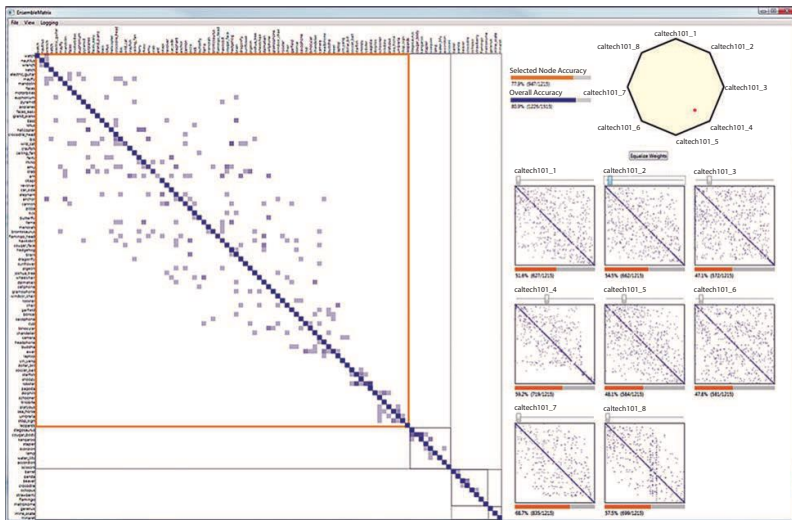
Best and worst matching
examples presentation

*=> Improved presentation
leads to better models*

Fogarty et al. (2008). CueFlik: interactive concept learning in image search. In *Proceedings of CHI* (pp. 29-38). ACM. [PDF](#)

EnsembleMatrix

- **Task:** classification
- Goal: create ensemble classifiers
- Show confusion matrix resulting from linear combination of classifiers

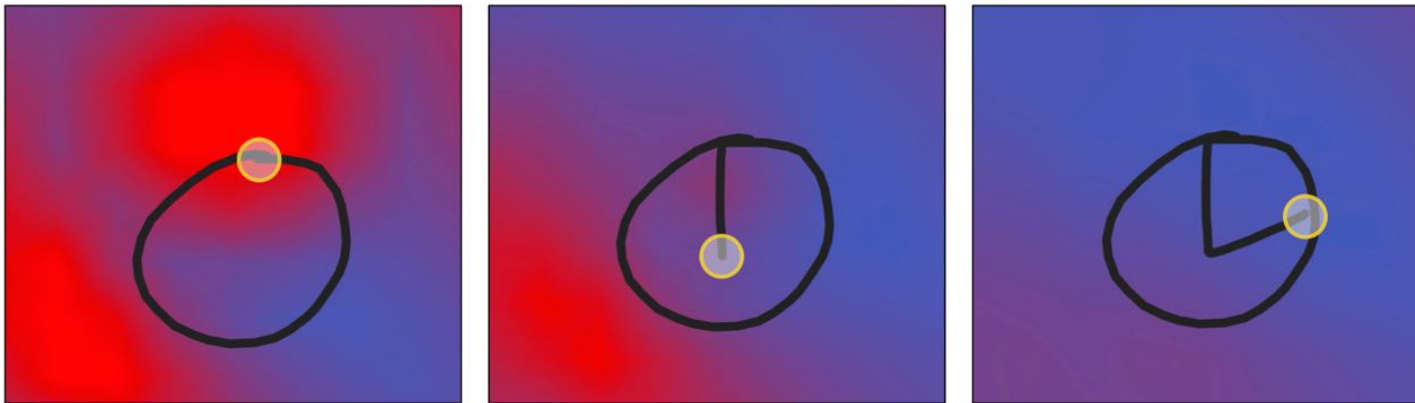


=> Thanks to their intuition and visual processing abilities, users quickly create state of the art ensemble classifiers

Talbot et al. (2009). EnsembleMatrix: interactive visualization to support machine learning with multiple classifiers. In Proceedings of CHI (pp. 1283-1292). ACM. [PDF](#)

Fieldward

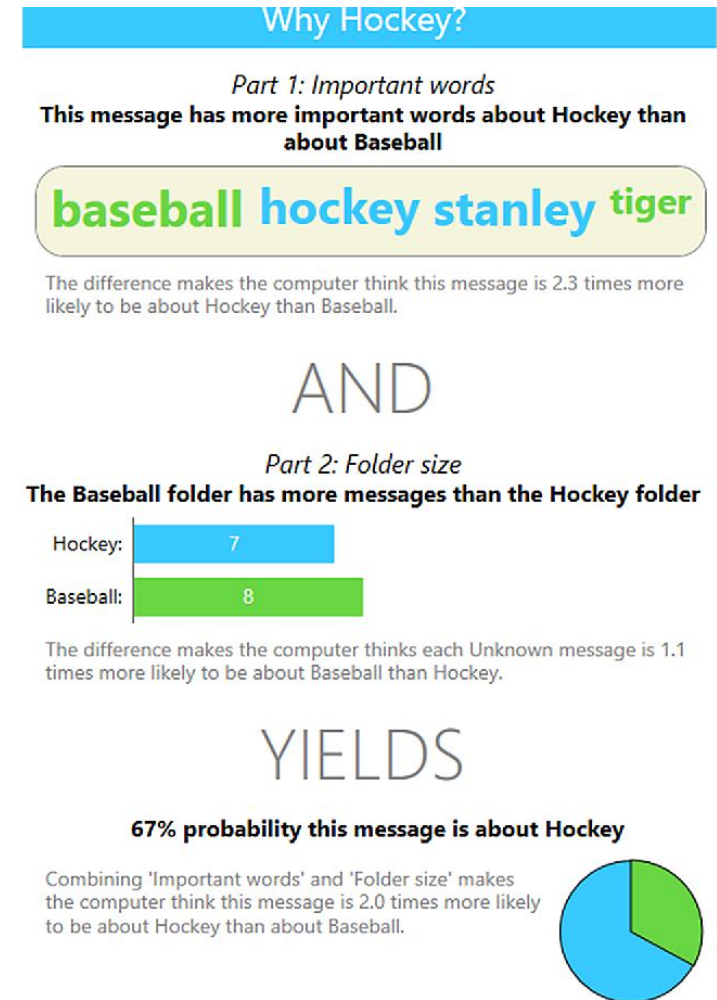
- **Task:** Gesture recognition
- Goal: help user create gestures that will be accurately recognized
- Visualize color gradients to reveal optimal directions for creating recognizable gestures



Malloch et al. (2017). *Fieldward and Pathward: Dynamic Guides for Defining Your Own Gestures*. In *Proceedings of CHI* (pp. 4266-4277). ACM. [PDF](#)

Elucidebug

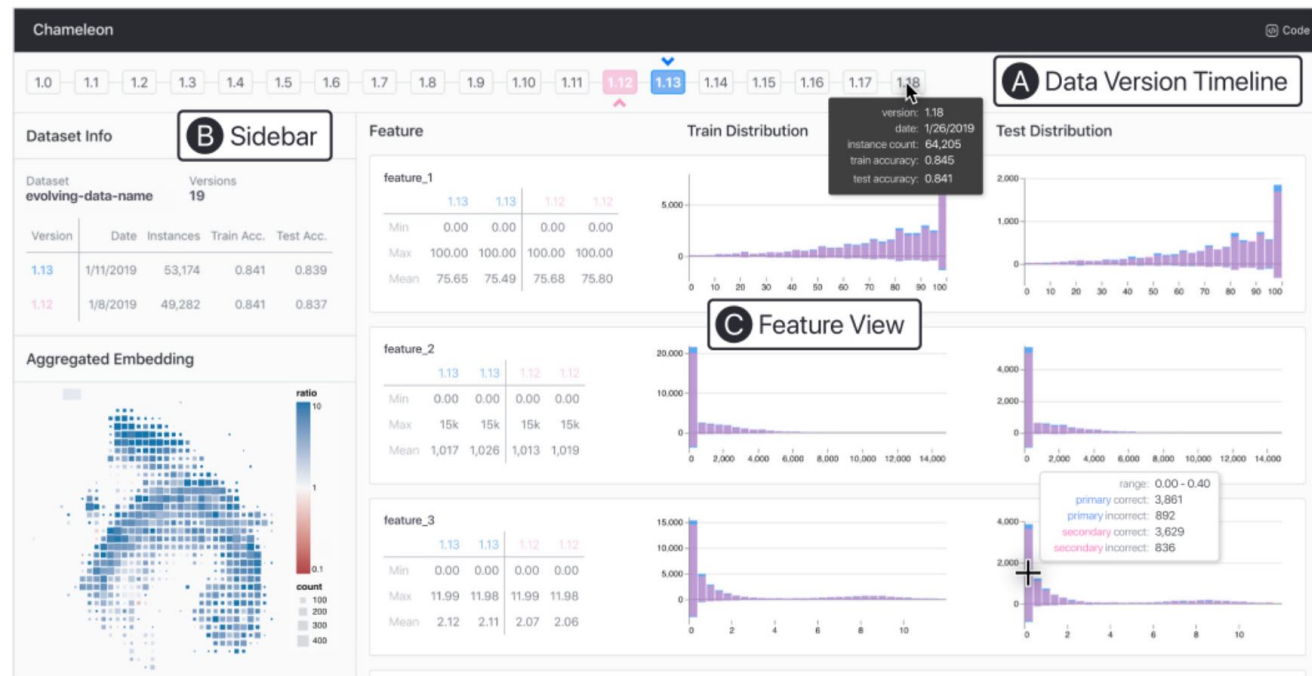
- **Task:** text analysis (spam filtering & classification)
- Goal: give users explanations about the algorithm's decisions
- Allow users to query the system about particular decisions
- Provides feedback about uncertainty



Kulesza et al. (2015). Principles of explanatory debugging to personalize interactive machine learning. ACM IUI. [PDF](#)

Chameleon

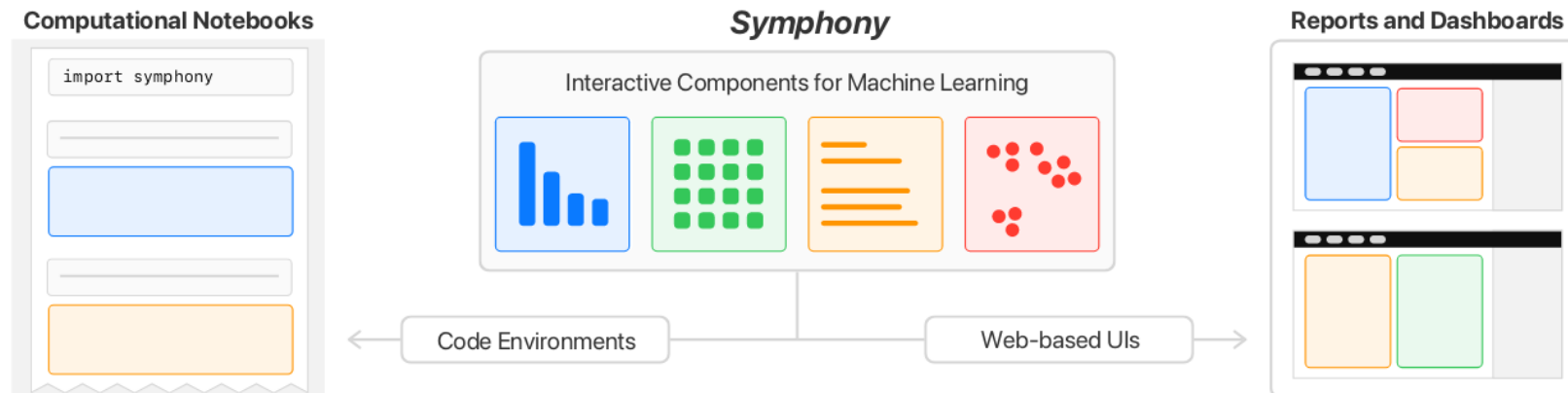
- **Task:** Classification
- Goal: ML experts can build more robust models
- Facilitate data iteration
- Think-aloud protocol



Hohman et al. (2020). Understanding and visualizing data iteration in machine learning. In Proceedings of CHI (pp. 1-13). [PDF](#)

Symphony

- **Task:** ML development
- Goal: Explore data and models, share interfaces and collaborate
- ML practitioners discover previously unknown issues (data duplicates, blind spots in models) + share insights with other stakeholders.



Bäuerle et al. (2022, April). Symphony: Composing interactive interfaces for machine learning. In *Proceedings of the 2022 CHI Conference on Human Factors in Computing Systems* (pp. 1-14).

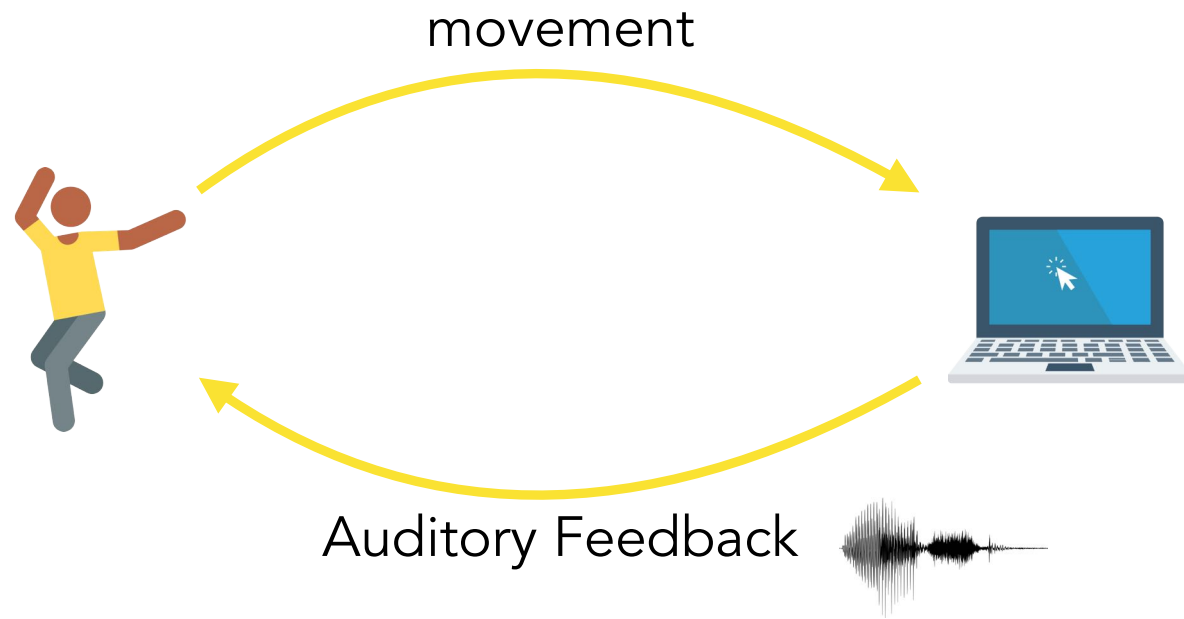
Designing by Demonstration

The Case of Motion-Sound Interactions

Ressources:

- Fiebrink et al. (2011). Human model evaluation in interactive supervised learning. In Proceedings of CHI (pp. 147-156). ACM.
- Françoise, J., & Bevilacqua, F. (2018). Motion-sound mapping through interaction: An approach to user-centered design of auditory feedback using machine learning. ACM Transactions on Interactive Intelligent Systems (TiiS), 8(2), 1-30.
- Fiebrink & Caramiaux (2018). The machine learning algorithm as creative musical tools. *Handbook of Algorithmic Music*

Movement-Sound Interactive Systems



real-time **continuous** and **multidimensional** interaction

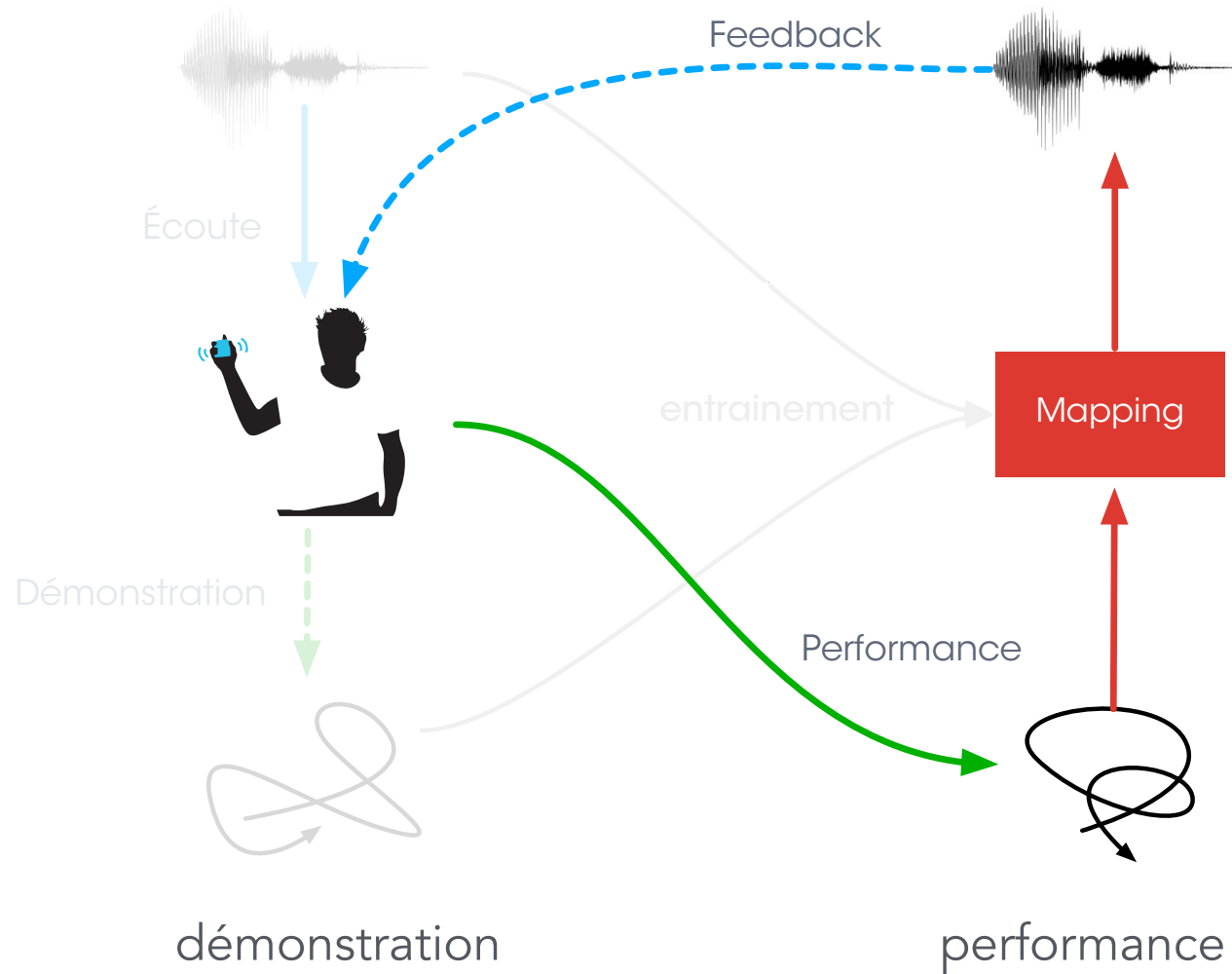
Designing movement-sound interactions



Designing through movement and interaction

Supporting appropriation

Mapping by Demonstration



Playing Sound Textures

Crafting hand control strategies by Demonstration

Jules Françoise, Riccardo Borghesi, Norbert Schnell, Frédéric Bevilacqua

from the SIGGRAPH'14 installation

MaD — Mapping by Demonstration for Continuous Sonification

Modular Musical Objects

IRCAM, NoDesign, Da Fact 2011



Designing with Machine Learning

- **Advantages**

- No need for a formal (mathematical) description of the problem
- Design using bodily knowledge
- Specification of (possibly complex) target gestures for sonification
- Individualised adaptation

- **But...**

- How do we make it work?

Designing with Machine Learning

- Process
 - The user imagines a vocabulary of gestures (+ sounds)
 - The user records demonstrations
 - The ML algorithm learns the mapping/classifier
 - It does **not** work (as well as we hope).
- Reasons for Failure
 - Technical factors (choice of sensor, features, model, and parameters)
 - Cognitive and sensorimotor factors (gesture design and execution)

ML: Specific Constraints

- User-centred approach in creating the database
 - Small data
 - linked to specific styles and approach
 - Possibly idiosyncratic to the user
 - unlabelled or inconsistent labelling
- Data evolving over time
 - learning processes from rehearsal to performance
 - Specific tasks and use cases

Choosing the right model

Models should be able to learn **fast** and from **few examples**

- => Difficult because expressive models often require a lot of data

- => Requires custom implementation and regularization

Understanding model properties is essential in Interaction Design

- => Models have different properties ($\neq$ design space)

- => How to explain models to novice users?

- => What to expose? (Learning behavior, parameters, ...)

Interactive Machine Teaching

What is a cat?



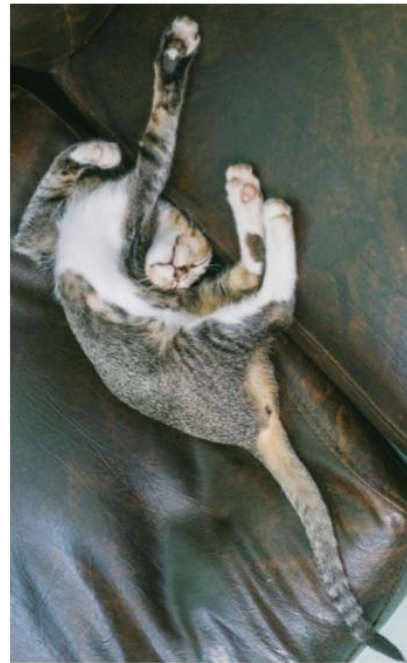
What is a cat?



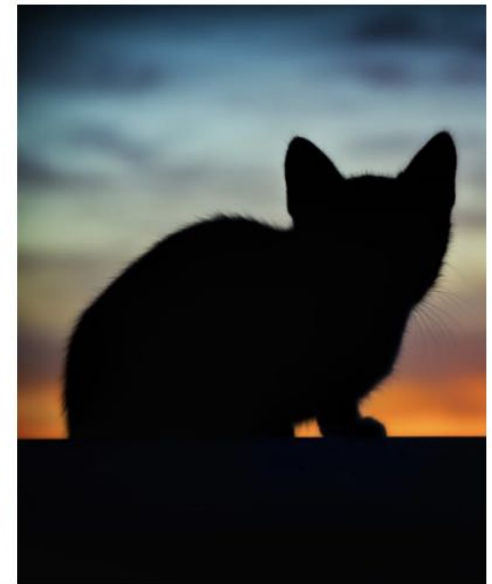
occlusion



diversity



deformation



lighting variations

Definition of Machine Teaching

"Machine Learning research aims at making the learner better by improving ML algorithms."

"Machine teaching research aims at making the teacher more productive at building machine learning models."

Engineering perspective (from Microsoft)

Motivation

- Related to problems in ML system development, deployment, maintenance
- Many issues with model building are not addressed with traditional ML
 - ex: concept evolution (teacher's underlying notion evolves over time)

Machine Teaching: Proposition for a new field of study

Definition of Interactive Machine Teaching

Interactive Machine teaching (IMT) is an IML process in which the human-in-the-loop takes the role of a **teacher**, and their goal is to create a machine-learned model.

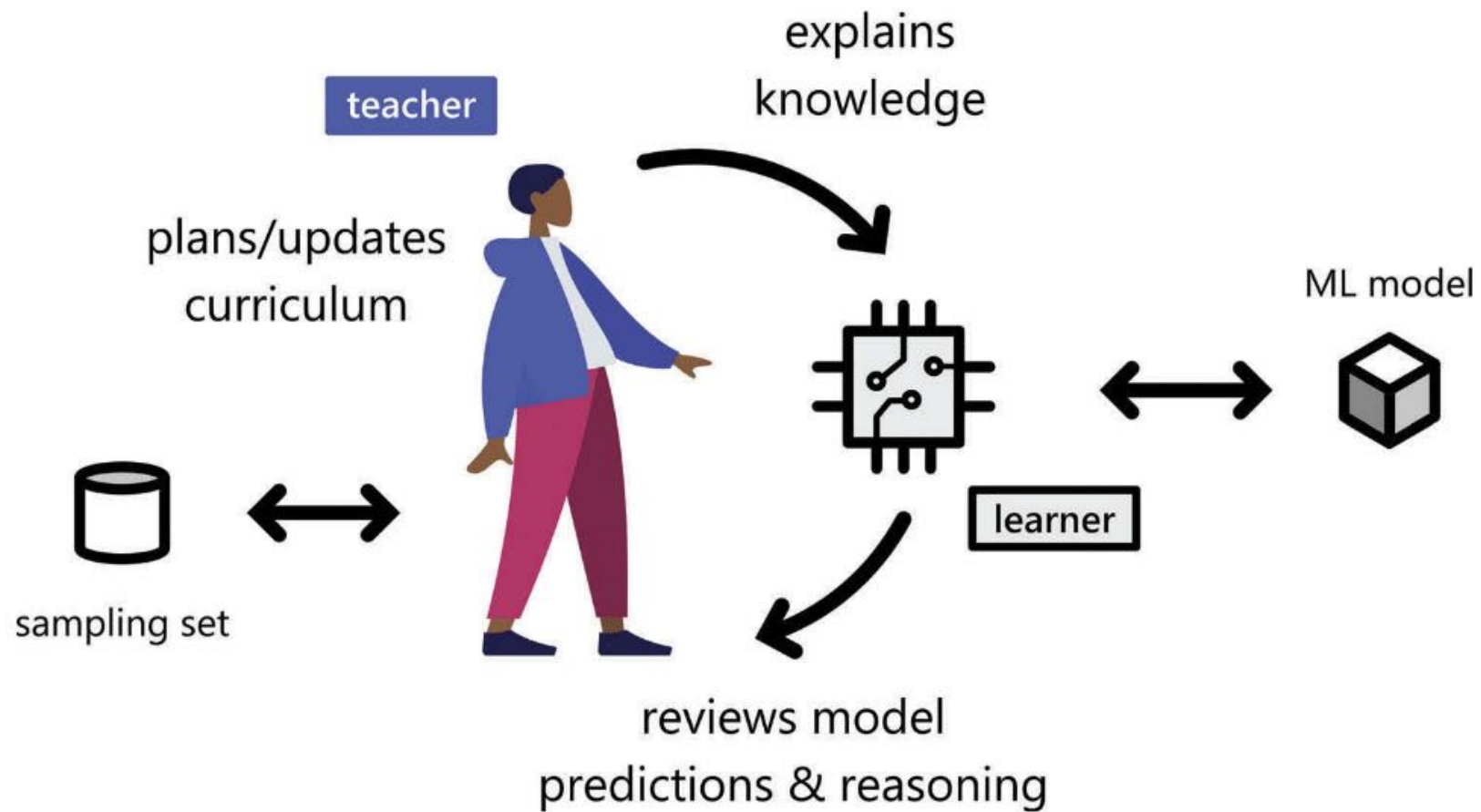
Model-building goal

- IML aims at Model Building or Task Completion

People as teachers

- In IML: variety of roles (providing labels, demonstrating, selecting features, ...)
- Variety of skills & ML expertise

The teaching process



Case study in sketch recognition

How do People Train a Machine? Strategies and (Mis)Understandings

TÉO SANCHEZ, Université Paris-Saclay, Inria, LISN, France

BAPTISTE CARAMIAUX, Université Paris-Saclay, CNRS, Inria, LISN & Sorbonne Université, ISIR, France

JULES FRANÇOISE, Université Paris-Saclay, CNRS, LISN, France

FRÉDÉRIC BEVILACQUA, STMS IRCAM-CNRS-Sorbonne Université, France

WENDY E. MACKAY, Université Paris-Saclay, CNRS, Inria, LISN, France

Machine learning systems became pervasive in modern interactive technology but provide users with little, if any, agency with respect to how their models are trained from data. In this paper, we are interested in the way novices handle learning algorithms, what they understand from their behavior and what strategy they may use to “make it work”. We developed a web-based sketch recognition algorithm based on Deep Neural Network (DNN), called *Marcelle-Sketch*, that end-users can train incrementally. We present an experimental study that investigate people’s strategies and (mis)understandings in a realistic algorithm-teaching task. Our study involved 12 participants who performed individual teaching sessions using a think-aloud protocol. Our results show that participants adopted heterogeneous strategies in which variability affected the model performances. We highlighted the importance of sketch sequencing, particularly at the early stage of the teaching task. We also found that users’ understanding is facilitated by simple operations on drawings, while confusions are caused by certain inherent properties of DNN. From these findings, we propose implications for design of IML systems dedicated to novices and discuss the socio-cultural aspect of this research.

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Conclusions

Interactive Machine Learning

=> *Bring the end-user in the development loop*

Principles and case study in movement-sound interaction

=> *Personalisation, immediacy, discoverability*

Interactive Machine Teaching

=> *Focus on the teacher*